23	**B**	Along PQ and RS, magnetic flux density *B* varies inversely proportional to distance *d*, such that $B = \dfrac{\mu_0 I}{2\pi d}$. At point O, equidistant from both X and Y, the magnetic flux density due to current in X is into the page and that due to current in Y is out of the page. Hence, the resultant magnetic flux density at O is zero.

Qn	Ans	Solution
24	D	Since the magnetic force on the ions provides the centripetal force to enable